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Enhanced Vision Transformer-based diagnosis of pediatric appendicitis using GAN-based data augmentation on ultrasound images

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ABSTRACT

Pediatric appendicitis is one of the most common surgical emergencies in children, yet its diagnosis remains challenging due to symptom variability and operator-dependent ultrasound interpretation. This study presents an automated diagnostic framework integrating synthetic ultrasound generation via StyleGAN2-ADA with a Swin Transformer classifier to support detection under data-scarce conditions. Using the Regensburg Pediatric Appendicitis dataset (578 patients; 1339 appendicitis and 382 non-appendicitis real images), we adopt a strict *patient-level* partitioning that prevents the leakage induced by image-level splits, train two class-specific generative models exclusively on training-set images, and benchmark the Swin Transformer against four convolutional baselines (ResNet50, DenseNet121, EfficientNet-B0, ConvNeXt-Tiny) using 5-fold patient-level cross-validation. Synthetic augmentation is evaluated at 0%, 50%, 100% and 200%, with significance assessed via bootstrap confidence intervals and McNemar's test; given the class imbalance we report AUC alongside balanced accuracy, sensitivity and specificity. Synthetic realism is examined through Kernel Inception Distance (KID ≈ 0.013), nearest-neighbor anti-memorization analysis, a real-vs-synthetic discriminability test, and expert review. The effect of augmentation is nuanced: AUC remains essentially unchanged (test up to 0.75; cross-validated ≈ 0.70), so it does not increase discrimination, but it rebalances the classifier—improving specificity, balanced accuracy and the Matthews correlation coefficient, with the best balance around 100% synthetic data. Our findings show that image-level evaluation markedly overestimates performance, that the hierarchical Vision Transformer is competitive with – rather than superior to – convolutional baselines, and they delineate when GAN-based augmentation does or does not benefit ultrasound classification under a leakage-free protocol.

1. Introduction

Pediatric appendicitis is one of the most common causes of acute abdominal pain in children and adolescents, with an estimated incidence of 70–100 cases per 100,000 children annually [1–3]. Despite its high prevalence, accurate and timely diagnosis remains a significant clinical challenge, particularly due to nonspecific clinical presentations and age-dependent symptom variability [4]. Undetected appendicitis can rapidly progress to severe complications such as perforation and peritonitis, leading to increased morbidity, prolonged hospital stays, and higher healthcare costs [5,6].

Ultrasound is widely adopted as the first-line imaging modality for pediatric appendicitis owing to its non-invasive nature and absence of ionizing radiation [2,7]. However, its diagnostic accuracy is highly dependent on the experience and skill of the examining clinician, image acquisition quality, and patient-specific conditions [8]. Even among experienced radiologists, substantial variability in ultrasound interpretation has been documented, resulting in false positives that may lead to unnecessary surgical interventions or false negatives that delay urgent treatment [9]. These limitations underscore the need for complementary diagnostic tools capable of reducing observer variability.

The application of artificial intelligence (AI) in medical imaging has shown considerable promise in addressing these challenges by automating image analysis and improving diagnostic consistency [4,10,11]. Nonetheless, the development of robust AI models for pediatric appendicitis detection is hindered by the limited availability of high-quality annotated datasets, a constraint frequently encountered in pediatric

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imaging research [12]. Ethical requirements, privacy regulations, and data-sharing restrictions further compound this problem [13].

Generative Adversarial Networks (GANs) [14] offer a promising solution to mitigate data scarcity by producing realistic synthetic medical images while preserving patient confidentiality [15,16]. The evolution of GAN architectures in medical imaging has been extensively documented, confirming their growing role as effective tools for data augmentation in clinical applications [17,18]. When combined with advanced classifiers, GAN-based augmentation has been shown to enhance model generalization without requiring extensive real-world datasets [19,20]. In parallel, Vision Transformers (ViTs) have demonstrated superior performance in capturing complex image patterns and long-range dependencies, making them well suited for medical imaging tasks including ultrasound-based diagnosis [21–24].

The primary objective of this work is to develop and evaluate an automated diagnostic framework for pediatric appendicitis that combines GAN-based synthetic ultrasound image generation with Vision Transformer-based classification. The proposed approach aims to enhance diagnostic accuracy, reduce variability in ultrasound interpretation, and address the limitations imposed by scarce pediatric imaging data.

The novelty of this study resides in the systematic integration of class-specific synthetic ultrasound image generation using StyleGAN2-ADA with a hierarchical Swin Transformer architecture for binary classification of pediatric appendicitis. Unlike conventional data augmentation techniques, the proposed pipeline leverages realistic GAN-generated images to expand the training set and evaluates the impact of four distinct real-to-synthetic data ratios on diagnostic performance. Furthermore, the realism and clinical plausibility of the synthetic images were validated by an independent physician specialized in ultrasound imaging, complemented by quantitative fidelity and anti-memorization analyses. Equally central to this work is the methodological rigor of the evaluation. All data partitions are performed at the *patient level* to prevent the data leakage that image-level splits introduce; the generative models and conventional augmentation are confined to the training set; and the Swin Transformer is benchmarked against strong convolutional baselines (ResNet50, DenseNet121, EfficientNet-B0 and ConvNeXt-Tiny) using patient-level cross-validation with bootstrap confidence intervals and McNemar's test. This protocol yields an honest, reproducible assessment of what GAN-based augmentation does – and does not – contribute, and quantifies how image-level evaluation can overestimate performance in this domain.

This study addresses three research questions: (1) under a leakage-free, patient-level protocol, to what extent does incorporating GAN-generated synthetic ultrasound images improve the diagnostic performance of the classifier? (2) what proportion of synthetic data relative to real data yields the best balance between sensitivity and specificity? and (3) does the hierarchical Vision Transformer outperform, or merely match, established convolutional baselines on this task?

The remainder of this paper is organized as follows. Section 2 reviews the related literature on medical image generation and classification. Section 3 describes the dataset, synthetic image generation process, and classification methodology. Section 4 presents the experimental results. Section 5 discusses the findings in context. Finally, Section 6 concludes the paper and outlines directions for future research.

2. Related work

The detection of pediatric appendicitis through medical imaging remains a complex diagnostic challenge that has motivated significant research at the intersection of generative modeling, data augmentation, and deep learning-based classification. This section reviews the relevant literature organized into three areas: data augmentation strategies in medical imaging, deep learning models for image classification, and AI-assisted diagnosis of appendicitis.

2.1. Data augmentation in medical imaging

Data augmentation encompasses techniques that artificially increase the size and diversity of a dataset so as to improve the generalization ability of machine learning models [25]. In medical imaging, augmentation is particularly valuable given the difficulty of obtaining large annotated datasets, especially in pediatric settings where privacy and ethical considerations further restrict data sharing [11,12].

Traditional augmentation methods – geometric transformations (rotation, flipping, scaling), noise injection, and contrast adjustments – have been shown to modestly improve performance in medical image classification [26]. However, their capacity to introduce fundamentally new visual patterns is inherently limited, particularly in settings characterized by class imbalance or rare conditions.

To overcome these limitations, GANs [14] have emerged as a powerful approach for synthesizing images that closely resemble real clinical scans. GAN-based augmentation has yielded performance improvements in tumor detection [18], lung disease diagnosis [16], and ultrasound-based classification [27,28]. Within the obstetric ultrasound domain, Lasala et al. [29] introduced FetalBrainAwareNet, a conditional GAN with anatomical regularization terms that synthesizes fetal head standard planes and provides a 3.2% improvement in classifier accuracy over training on real acquisitions alone, illustrating the value of class-specific generative models for ultrasound augmentation. A recent scientometric analysis spanning the period 2016–2025 confirmed the sustained and accelerating trend of GAN adoption in medical imaging research, identifying data augmentation as one of the dominant application areas [17]. In parallel, studies on dermatological imaging have demonstrated that GAN-based data augmentation significantly improves binary classification accuracy for skin lesion detection when combined with deep learning classifiers [20]. Recent work has also emphasized the role of synthetic data in mitigating scarcity while preserving patient privacy within emerging regulatory frameworks such as the European Health Data Space [13]. Nonetheless, challenges including mode collapse and the potential introduction of unrealistic artifacts require careful clinical expert validation [19].

2.2. Deep learning models for medical image classification

Medical image classification has historically relied on Convolutional Neural Networks (CNNs), which automatically learn hierarchical features from image data [11]. CNNs have been successfully applied to appendicitis diagnosis from ultrasound images [9,30], although they exhibit well-known limitations including dependence on large labeled datasets, constrained receptive fields, and susceptibility to overfitting in small-data regimes. Retrospective clinical studies have further shown that while CNN-based models achieve high sensitivity, their generalization across institutions and imaging protocols remains limited [31].

Vision Transformers (ViTs) address several of these shortcomings by processing images as sequences of patches and using self-attention mechanisms to model long-range dependencies [23]. Multiple reviews have reported the superior capacity of transformer-based architectures to capture global contextual information in medical images across diverse modalities, including ultrasound [21,22,32,33]. Among ViT variants, the Swin Transformer [34] has attracted particular attention owing to its hierarchical architecture and shifted window self-attention, which substantially reduce computational cost while preserving classification performance. Beyond two-dimensional inputs, fully transformer-based pipelines have also been proposed for volumetric classification: Oghbaie et al. [35] report a variable-length feature aggregator transformer (VLFATRollout) that classifies retinal optical coherence tomography volumes and improves balanced accuracy by 5.47% over leading convolutional baselines. Recent benchmarking studies comparing ViTs with CNNs for medical image classification have similarly confirmed that transformer-based models, including Swin Transformers, achieve

competitive or superior results across standard benchmarks [24,36]. Beyond classification, the broader role of attention-based architectures in image analysis continues to expand: recent lightweight transformer designs incorporating edge-aware feature fusion have been shown to improve structural feature representation and cross-domain robustness while reducing computational cost [37], motivating the use of a hierarchical, locality-aware transformer such as the Swin Transformer for fine-grained ultrasound patterns. Within *Knowledge-Based Systems*, transformer- and attention-based pipelines have likewise advanced medical image analysis: Iqbal and Sharif [38] proposed a Swin Transformer network (BTS-ST) for segmentation and classification of multimodality breast cancer images, Wu et al. [39] combined a Transformer and a CNN (CTransCNN) for multilabel medical image classification, and Zhu et al. [40] introduced an ensemble of bio-inspired evolutionary DenseNets (MEEDNets) for medical image classification, underscoring the relevance of both transformer and convolutional designs to the present task.

2.3. AI-assisted diagnosis of appendicitis

The application of AI to appendicitis diagnosis has accelerated in recent years. A comprehensive review by Akbulut et al. [4] surveyed both machine learning and deep learning applications, documenting incremental improvements in diagnostic accuracy but noting persistent challenges related to data availability, generalization, and clinical adoption. In the pediatric context specifically, Marcinkevics et al. [8] developed interpretable and intervenable machine learning models using the Regensburg dataset, combining ultrasonography with structured clinical data. Navaei et al. [10] integrated clinical, laboratory, and imaging data for pediatric diagnosis. Tan et al. [31] retrospectively developed an AI model combining ultrasound and clinical features for pediatric appendicitis differentiation. Kendall et al. [41] explored feature selection techniques in the pediatric appendicitis context. More recently, Hosseinabadi and Sharifi [42] addressed pediatric appendicitis detection directly from ultrasound images, while Ogut et al. [43] proposed a hybrid CNN–ViT architecture with quantum feature maps for appendicitis classification. Outside appendicitis but illustrative of the broader use of multi-task transformer pipelines for ultrasound diagnosis, Gan et al. [44] proposed SML-Net, a semi-supervised multi-task framework that jointly segments and classifies carotid plaques, achieving 86.59% classification accuracy on ultrasound data and exemplifying the clinical utility of joint-learning architectures on ultrasound imaging.

Table 1 provides a comparative overview of the most relevant related approaches together with the framework proposed in this study. The key differentiating aspect of the present work lies in the joint use of class-specific GAN-based data augmentation and a hierarchical Vision Transformer classifier for pediatric appendicitis detection from ultrasound images, a combination that remains unexplored in the existing literature.

Table 1
Comparative overview of related approaches for AI-based appendicitis diagnosis.

	Year	Approach	Data modality	GAN aug.
Yilmaz et al. [30]	2021	CNN classification	Ultrasound	No
Marcinkevics et al. [8]	2024	Interpretable ML models	US + clinical	No
Hariri et al. [9]	2024	LesionScanNet (CNN)	Ultrasound	No
Tan et al. [31]	2025	AI + clinical features	US + clinical	No
Navaei et al. [10]	2025	ML + multimodal fusion	Clinical + lab + US	No
Kendall et al. [41]	2025	ML + feature selection	Clinical	No
Hosseinabadi & Sharifi [42]	2025	ML detection	Ultrasound	No
Ogut et al. [43]	2026	CNN–ViT–quantum hybrid	Medical images	No
This work	2026	Swin-T + StyleGAN2-ADA	Ultrasound (1721)	Yes

3. Materials and methods

This section describes the dataset, the synthetic image generation pipeline, the classification architecture, and the experimental configuration employed in this study.

3.1. Dataset

This study utilizes the Regensburg Pediatric Appendicitis Dataset [12], a publicly available collection of grayscale ultrasound images and structured clinical data gathered between 2016 and 2021 at the Children's Hospital St. Hedwig in Regensburg, Germany. The dataset comprises ultrasound scans accompanied by demographic information, presenting symptoms, laboratory results, and both presumptive clinical diagnoses (based on initial assessment) and final confirmed diagnoses (established through surgical findings or imaging-confirmed resolution). Only cases with a confirmed final diagnosis were retained for model training and evaluation.

Table 2
Patient-level dataset accounting (official held-out test split). Conventional augmentation and GAN-generated images are added only within the training set; validation and test contain only real, unaugmented images.

Subset	Patients	Appendicitis img.	Non-app. img.
Training	463	1062	298
Validation	58	112	57
Test (official)	57	165	27
Total (real)	**578**	**1,339**	**382**
Excluded (missing label)	–	370 images	

3.1.1. Dataset composition and class balancing

The original dataset contains 2091 ultrasound images, of which 1721 are associated with a confirmed final diagnosis (1339 from patients with confirmed appendicitis and 382 from patients without appendicitis); the remaining 370 images were excluded due to missing label references in the clinical metadata. Because most patients contribute several views (between 1 and 15 images each), all partitions in this study are performed strictly at the *patient level*: no patient contributes images to more than one of the training, validation or test sets. This eliminates the inter-split data leakage that an image-level partition would otherwise introduce, and which can substantially inflate measured performance. Table 2 summarizes the resulting accounting: 578 patients with usable, confirmed-diagnosis images (382 appendicitis and 196 non-appendicitis), comprising 1721 real ultrasound images. The pronounced class imbalance (approximately 3.5:1) is addressed *only within the training set*, after partitioning, through class-balanced sampling and conventional augmentation applied on-the-fly during training (random rotations, horizontal flips, and brightness/contrast adjustments). Consequently, the validation and test sets consist exclusively of pristine, unaugmented, real ultrasound images, and no augmented copies of a training image can appear in evaluation.

Note that the demographic characterization (Section 3.1.3) and the presumptive-versus-confirmed analysis (Section 3.1.4) refer to the full Regensburg cohort with available clinical records (755 patients), whereas all generative-model training, classifier training and evaluation use exclusively the subset of 578 patients with a confirmed final diagnosis and usable ultrasound images (Table 2).

3.1.2. Image characteristics and preprocessing

The images in the dataset exhibit notable heterogeneity in format and resolution. The predominant format is BMP (98.4%, 2064 images), followed by PNG (1.5%, 31 images) and JPG (0.1%, 2 images). All images were converted to PNG format for compatibility with deep

Table 3
Demographic characteristics of the full patient cohort (755 patients).

Characteristic	Description
Total patients	754
Gender (male/female)	52.4%/47.6%
Confirmed appendicitis (male/female)	254/188 (57.4%/42.6%)
Non-appendicitis (male/female)	141/171 (45.2%/54.8%)
Age range	0–18 years
Peak age group	10–15 years
Height range (majority)	140–160 cm
Weight range (majority)	30–40 kg

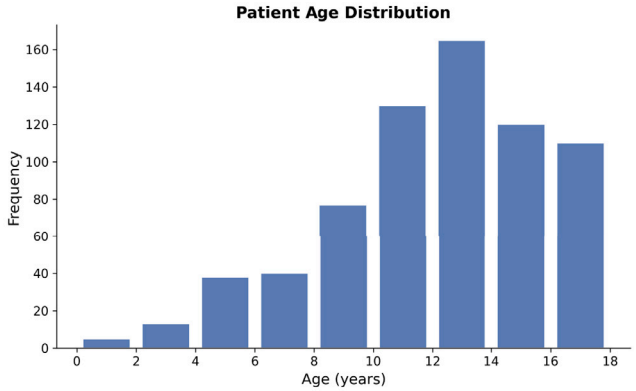


Fig. 1. Distribution of patient ages in the dataset. The peak incidence is observed in the 10–15 year age group, consistent with the epidemiology of pediatric appendicitis [45].

learning frameworks. Spatial resolutions varied considerably, with the most frequent resolution being 716 × 537 pixels (47.7% of images), followed by other dimensions such as 799 × 581 and 800 × 584. To ensure input uniformity, all images were resized to 256 × 256 pixels for GAN training and subsequently to 224 × 224 pixels for the Swin Transformer classifier.

3.1.3. Patient demographics

Table 3 summarizes the demographic characteristics of the patient cohort. The dataset exhibits a balanced gender distribution (52.4% male, 47.6% female), minimizing potential gender-related biases. The age distribution is consistent with epidemiological studies of pediatric appendicitis, which report peak incidence in the 10–15 year range [3, 45]. Height and weight distributions correspond to expected growth standards for the represented age groups.

A slight predominance of confirmed appendicitis among male patients (57.4% vs. 42.6% for females) is consistent with clinical literature reporting higher appendicitis incidence in males [45]. Fig. 1 illustrates the patient age distribution, which peaks in the 10–15 year range, aligning with the known epidemiological profile of pediatric appendicitis. These demographic patterns are representative of the target population and provide a solid foundation for model development.

3.1.4. Diagnostic challenge

Table 4 summarizes the distribution of presumptive versus confirmed diagnoses, illustrating the diagnostic challenge that motivates this work. The relatively high recall for appendicitis in the presumptive stage (0.98) combined with low precision (0.65) reflects a conservative clinical strategy that prioritizes sensitivity over specificity, leading to a substantial false-positive rate in initial assessments.

Table 4
Diagnostic classification report: presumptive versus confirmed diagnosis.

	Precision	Recall	F1-Score	Support
No appendicitis	0.91	0.25	0.39	312
Appendicitis	0.65	0.98	0.78	443
Macro average	0.78	0.62	0.59	755
Weighted average	0.76	0.68	0.62	755

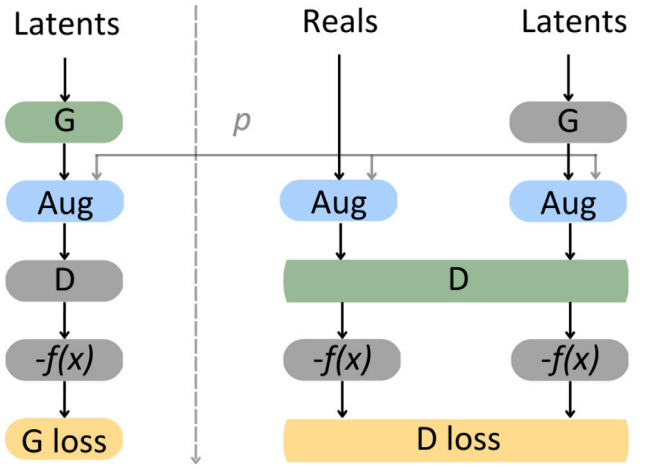


Fig. 2. StyleGAN2-ADA architecture. The generator produces synthetic ultrasound images from latent vectors, while the discriminator distinguishes between real and generated images.
Source: Adapted from [47].

3.2. Synthetic image generation using StyleGAN2-ADA

To address data scarcity, a StyleGAN2 architecture with Adaptive Discriminator Augmentation (ADA) [46] was employed to generate synthetic ultrasound images. StyleGAN2-ADA was selected for its ability to produce high-fidelity images even with limited training samples, its fine-grained control over generated image features, and its ADA mechanism that dynamically adjusts augmentations to prevent discriminator overfitting.

Two separate generative models were trained using *only the training-set real images* of each class (1062 appendicitis and 298 non-appendicitis images; see Table 2). No validation or test images were used to train the generators, and no conventionally-augmented images were fed to them: the ADA mechanism alone provides the augmentation needed in this low-data regime, as recommended for StyleGAN2-ADA. This class-specific strategy ensures that the resulting synthetic dataset preserves the balanced distribution of pathological and non-pathological cases, and it follows the same anatomically aware, class-conditional generation principle adopted in recent GAN-based ultrasound synthesis pipelines [29]. Fig. 2 presents a schematic overview of the StyleGAN2-ADA architecture.

Both models share an identical training configuration optimized for stability and image quality. The generator operates on a latent space of dimension $z = 512$, mapped to an intermediate style space $\mathcal{W}$ through an 8-layer mapping network. The discriminator employs convolutional blocks with adaptive normalization to evaluate the generated images. Both networks are optimized using the Adam optimizer with $\beta_1 = 0$ and $\beta_2 = 0.99$. Input images were resized to 256 × 256 pixels and converted to TFRecords format for efficient I/O during training. Training was conducted for 1000 kimg (thousands of generated images), with the ADA mechanism dynamically applying geometric (rotations, scaling, translations) and photometric (color adjustments) augmentations to prevent discriminator overfitting. All training was performed on Google Colab using NVIDIA A100 GPUs. For each class, the model checkpoint

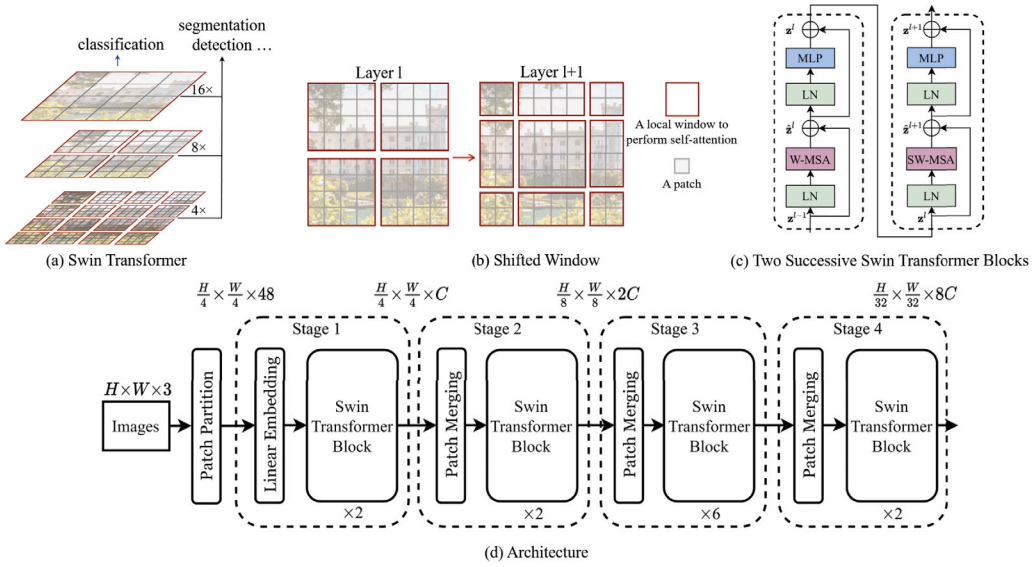


Fig. 3. Swin Transformer architecture: patch partitioning, shifted window attention, and hierarchical feature extraction. *Source:* Adapted from [34].

with the lowest Kernel Inception Distance (KID) was selected as the final model.

The quality of the generated images was assessed using the Kernel Inception Distance (KID) [48], an unbiased estimator of the squared Maximum Mean Discrepancy (MMD) between real and synthetic Inception-v3 feature distributions using a polynomial kernel. Unlike the Fréchet Inception Distance, the KID is unbiased and well behaved with only a few hundred samples per class, which makes it appropriate for our low-data regime. Lower KID values indicate greater similarity between real and synthetic images:

$$\mathrm{KID} = \frac{1}{m(m-1)} \sum_{i \neq j} k(x_i, x_j) + \frac{1}{n(n-1)} \sum_{i \neq j} k(y_i, y_j) - \frac{2}{mn} \sum_{i,j} k(x_i, y_j), \quad (1)$$

where x_i and y_j are Inception features of real and generated images, and $k(a, b) = \left(\frac{1}{d} a^\top b + 1\right)^3$ is the polynomial kernel.

3.3. Classification using swin transformer

Image classification was performed using the Swin Transformer [34], a hierarchical Vision Transformer designed for efficient multi-scale feature extraction. Unlike standard ViTs, which operate on fixed-size patches throughout, the Swin Transformer progressively merges neighboring patches to build a multi-scale feature pyramid while constraining self-attention computation to local windows. Shifted window attention across successive layers enables the capture of cross-window dependencies and global context, achieving a favorable balance between computational efficiency and representational capacity. The architecture is illustrated in Fig. 3.

The specific variant used in this study is `swin_tiny_patch4_window7_224`, pre-trained on ImageNet-1K and fine-tuned for binary appendicitis classification. This model was implemented using the `timm` library in PyTorch. The output layer was replaced with a single fully connected layer for binary classification (appendicitis vs. non-appendicitis), enabling transfer learning from general-purpose image features to the medical domain. This variant offers a practical balance between computational cost and classification accuracy, making it suitable for deployment in resource-constrained clinical environments.

Table 5
Training-set sizes for each augmentation configuration (patient-level split). Synthetic images are added only to the training set, in proportion to the real images of each class; the validation (169) and test (192) sets are fixed and contain only real, unaugmented images.

Configuration	% Synthetic	Appendicitis (train)	No appendicitis (train)
Model 1	0%	1062	298
Model 2	50%	1593	447
Model 3	100%	2124	596
Model 4	200%	3186	894

3.4. Experimental configuration

3.4.1. Data preprocessing

All ultrasound images were converted from BMP to PNG format for compatibility with deep learning frameworks. Pixel intensity values were normalized using histogram equalization and Z-score standardization to ensure uniform contrast and brightness. Images were resized to 224 × 224 pixels, matching the input requirements of the Swin Transformer. Data were partitioned strictly at the patient level (Section 3.1). For the synthetic-augmentation study we use the dataset authors' official held-out test set and a stratified patient-level training/validation split (Table 2); for the architecture comparison we additionally use 5-fold patient-level cross-validation. No patient appears in more than one partition in any experiment.

3.4.2. Training configurations

To investigate the effect of synthetic data on classification performance, four training configurations were evaluated, as summarized in Table 5. Model 1 uses only real images with conventional data augmentation (random rotations, horizontal flips, brightness and contrast adjustments). Models 2–4 progressively increase the proportion of GAN-generated images at 50%, 100%, and 200% of the real training set, respectively. Critically, the validation and test sets remain identical across all configurations and consist exclusively of real ultrasound images, ensuring a fair comparison.

Convolutional baselines. To assess whether a hierarchical Vision Transformer is necessary on this dataset, the Swin Transformer is benchmarked against four widely used convolutional architectures – ResNet50, DenseNet121, EfficientNet-B0 and ConvNeXt-Tiny – trained

under identical preprocessing, partitions, hyperparameters and augmentation.

Evaluation protocol and statistics. Two complementary protocols are used. (i) A fixed patient-level split with the dataset authors' official held-out test set supports the synthetic-augmentation study (0/50/100/200%). (ii) A 5-fold patient-level cross-validation on real data supports the architecture comparison, reporting mean ± standard deviation across folds. Because the test set is class-imbalanced (≈86% appendicitis), the area under the ROC curve (AUC) is reported as the primary, threshold-independent metric, complemented by balanced accuracy, sensitivity and specificity at an operating point selected on the validation set (Youden's J). Statistical significance is assessed with 95% bootstrap confidence intervals and McNemar's paired test.

3.4.3. Training hyperparameters

All models were trained using the AdamW optimizer with a learning rate of 1×10^{-4} and weight decay of 1×10^{-5}. The batch size was set to 32 with mixed-precision training (using PyTorch's `GradScaler`) to optimize GPU memory usage. Training proceeded for up to 100 epochs, with the best model selected based on the highest validation accuracy observed during training. The Cross-Entropy loss function was used for binary classification. Standard data augmentation – random rotations (±10°), horizontal flips, brightness and contrast adjustments (±20%), and normalization to the range $[-1, 1]$ using mean 0.5 and standard deviation 0.5 – was applied consistently to the training set across all configurations. All classifier training was conducted on Google Colab using NVIDIA A100 GPUs.

4. Results

This section reports the quality assessment of the synthetic images generated by StyleGAN2-ADA, followed by the classification performance of the Swin Transformer across the four training configurations.

All results reported below were obtained under the corrected patient-level protocol (Section 3). The held-out test set comprises 192 real, unaugmented ultrasound images (165 appendicitis, 27 non-appendicitis); given this class imbalance, AUC is used as the primary discrimination metric and balanced accuracy, sensitivity, specificity and the Matthews correlation coefficient (MCC) are reported at an operating point selected on the validation set (Youden's J).

4.1. Quality of synthetic images

The fidelity of the synthetic images was assessed with three complementary analyses: a nearest-neighbor (anti-memorization) test, an automated real-vs-synthetic discriminability test, and an expert qualitative review. We note that the Fréchet Inception Distance (FID) reported in the original submission proved numerically unreliable under the corrected protocol: with only a few hundred real training images per class, the 2048-dimensional Inception feature covariance is severely rank-deficient and the FID estimator becomes unstable (yielding non-physical values). As a robust, small-sample alternative we therefore report the Kernel Inception Distance (KID), an unbiased estimator that is well behaved in low-data regimes; the trained generators attain low KID values of 0.013 for appendicitis and 0.013 for non-appendicitis (precisely 0.0132 and 0.0128), indicating good distributional fidelity. Notably, this small aggregate distance coexists with near-perfect machine discriminability (below), reflecting subtle, systematic "GAN fingerprints" rather than a large distributional shift. We complement KID with the analyses below.

No memorization. For a random sample of 200 synthetic images per class, we retrieved the nearest real training image in Inception feature space (cosine distance). The minimum distances were clearly above zero (appendicitis: mean 0.082, min 0.040; non-appendicitis: mean 0.089, min 0.033) and the closest pairs are visually distinct (Fig. 4), indicating that the generators synthesize novel variations rather than reproducing training images.

Table 6
Swin Transformer on the held-out test set (192 real images) by synthetic-augmentation ratio. AUC is threshold-independent; balanced accuracy, MCC, sensitivity and specificity are at the validation-selected operating point. 95% bootstrap confidence intervals in brackets.

Config.	AUC	Bal. Acc.	MCC	Sens.	Spec.
0%	0.747 [0.64,0.84]	0.619 [0.53,0.72]	0.221	0.867	0.370
50%	0.749 [0.64,0.85]	0.670 [0.58,0.76]	0.240	0.636	0.704
100%	0.705 [0.58,0.82]	**0.722** [0.63,0.81]	**0.316**	0.667	**0.778**
200%	0.744 [0.64,0.84]	0.707 [0.61,0.80]	0.295	0.673	0.741

Machine discriminability. Conversely, a ResNet-18 classifier trained to separate real from synthetic images did so almost perfectly (accuracy and AUC ≈ 1.0 on a held-out set of 659 images). This is consistent with the well-documented existence of detectable "GAN fingerprints": although the synthetic images are visually plausible, they retain machine-detectable distributional differences from real ultrasound. As shown in Section 4.2, this residual gap helps explain why synthetic augmentation does not improve discrimination (AUC), even when the generated images are visually convincing.

Fig. 5 presents representative examples of real and synthetic ultrasound images for both classes, illustrating the visual fidelity achieved by the generative models.

In addition to quantitative and visual analysis, a qualitative clinical assessment was performed by an independent physician certified in ultrasound imaging and specialized in prenatal diagnosis of congenital malformations. The specialist reviewed a representative sample of synthetic images and concluded that they exhibit a high degree of visual realism with adequate anatomical coherence, without evident artifacts that could hinder clinical interpretation. From a clinical perspective, the synthetic images adequately reproduce the relevant ultrasound features of the appendix and surrounding tissues, supporting their suitability as training data for AI-based diagnostic models. We note that a formal blinded Visual Turing Test by the clinician could not be completed within the revision period; the nearest-neighbor and machine-discriminability analyses above were added to provide objective, reproducible fidelity evidence, and the absence of a clinician-based Visual Turing Test is acknowledged as a limitation.

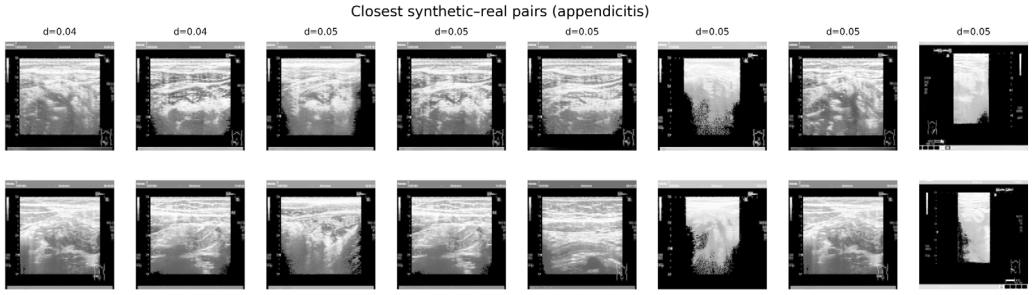
4.2. Classification performance

Table 6 reports the Swin Transformer across the four synthetic-augmentation configurations on the held-out test set, and Table 7 the architecture comparison under 5-fold patient-level cross-validation.

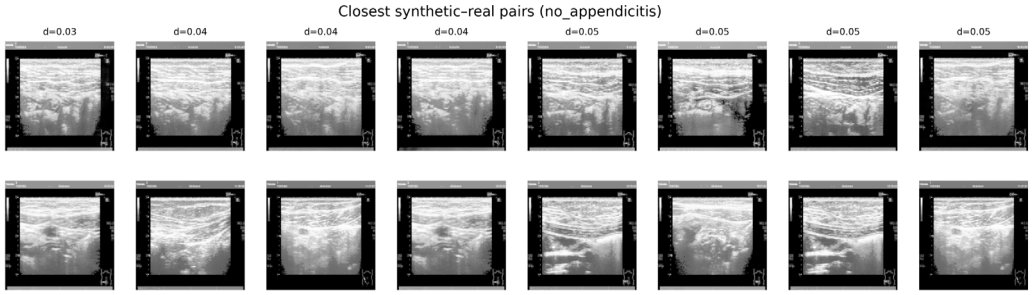
On the held-out test set, the real-only Swin model attains an AUC of 0.747. Adding synthetic data leaves the AUC essentially unchanged (0.705–0.749 across configurations, with broadly overlapping confidence intervals), indicating that GAN-based augmentation does *not* improve the model's intrinsic discriminative ability. What synthetic augmentation does change is the *balance* of the predictions: balanced accuracy rises from 0.619 (0%) to a peak of 0.722 at 100% synthetic, MCC from 0.221 to 0.316, and specificity from 0.370 to 0.778, at the cost of sensitivity (0.867 to 0.667). The most balanced operating regime is obtained around 100% synthetic data.

We caution that, on this imbalanced test set (≈86% appendicitis), overall accuracy and a McNemar test on raw correctness favor the real-only model (Swin 0% vs. 50%: $p < 0.001$) simply because it predicts the majority class more often; the threshold-independent AUC and the balanced metrics provide a fairer assessment and are emphasized here.

Under cross-validation, all five architectures perform comparably (AUC 0.671–0.711, with overlapping standard deviations), and pairwise McNemar tests reveal no statistically significant difference between the Swin Transformer and the convolutional baselines (all $p > 0.07$). The hierarchical Vision Transformer is therefore *competitive with – but not statistically superior to –* strong CNN baselines on this dataset,



(a) Appendicitis.



(b) Non-appendicitis.

Fig. 4. Closest synthetic–real pairs, stacked by class. Within each panel the top row shows synthetic images and the bottom row their nearest real training image (Inception cosine distance). Small but non-zero distances and visually distinct pairs indicate diversity without memorization.

Table 7
Architecture comparison under 5-fold patient-level cross-validation (real data only). Mean ± standard deviation across folds.

Architecture	AUC	Bal. Acc.	MCC
Swin-Tiny (ours)	0.700 ± 0.043	0.645 ± 0.042	0.249 ± 0.073
ResNet50	0.709 ± 0.066	0.659 ± 0.049	0.281 ± 0.091
DenseNet121	0.702 ± 0.053	0.640 ± 0.050	0.245 ± 0.087
EfficientNet-B0	0.671 ± 0.042	0.595 ± 0.022	0.176 ± 0.055
ConvNeXt-Tiny	0.711 ± 0.033	0.640 ± 0.041	0.252 ± 0.070

while attaining the highest single-split test AUC (0.747). These findings directly temper the original claim of transformer superiority.

Fig. 6 shows the confusion matrices of the Swin Transformer for the four configurations at the validation-selected operating point.

The matrices make the rebalancing explicit: the real-only model (0%) correctly identifies only 10 of 27 non-appendicitis cases, whereas synthetic augmentation raises this to 19–21 (50%–200%), at the cost of more missed appendicitis cases. This is the operating-point shift already reflected in the higher specificity and balanced accuracy of Table 6; the underlying discrimination (AUC) remains unchanged.

Fig. 7 shows Grad-CAM/attention maps for the Swin Transformer and a representative CNN (ResNet50) on test images. The salient regions concentrate on the appendiceal area and surrounding tissues rather than on peripheral artifacts, providing qualitative evidence that the models attend to clinically meaningful structures.

5. Discussion

5.1. Effect of synthetic data proportion

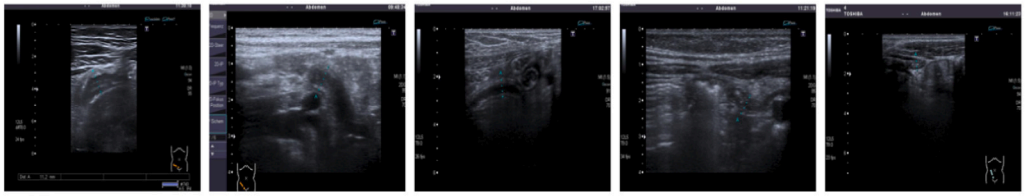
The results reveal a clear dissociation between *discrimination* and *operating balance*. Increasing the proportion of synthetic data (50%–200%) leaves the AUC essentially unchanged (Table 6), indicating that GAN-based augmentation does not enhance the model's intrinsic ability to separate the two classes. Instead, it shifts the classifier toward more balanced predictions: balanced accuracy, specificity and MCC improve and peak around 100% synthetic data, at the expense of sensitivity. This contrasts sharply with the inflated, seemingly monotonic "accuracy" gains reported in our original image-level analysis, which we now attribute largely to data leakage and to the high prevalence of the majority class rather than to a genuine effect of augmentation.

A plausible explanation is given by the discriminability analysis of Section 4.1: although the synthetic images are visually realistic and non-memorized, a classifier separates them from real images almost perfectly, i.e. they carry detectable distributional differences. Adding such data enriches the minority class enough to rebalance the decision boundary, but it does not inject genuinely new discriminative information about real ultrasound, which is consistent with the flat AUC. This nuance – augmentation as a rebalancing mechanism rather than a source of discriminative power – is, we believe, a more honest and useful conclusion than a headline accuracy gain.

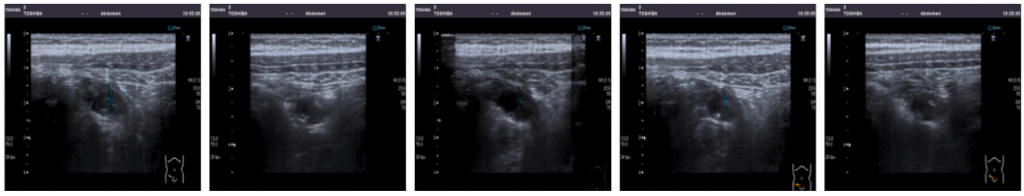
5.2. Comparison with prior work

The proposed framework offers two advantages over existing approaches for AI-based appendicitis diagnosis. First, unlike recent works that rely on multimodal data fusion (combining imaging with laboratory results and clinical features) [8,10,31], the present approach operates exclusively on ultrasound images, which simplifies deployment and avoids the need for structured clinical data integration. Second, the combination of GAN-based data augmentation with a Vision Transformer classifier has not been previously applied to the pediatric appendicitis domain, as confirmed by the comparative analysis presented in Table 1.

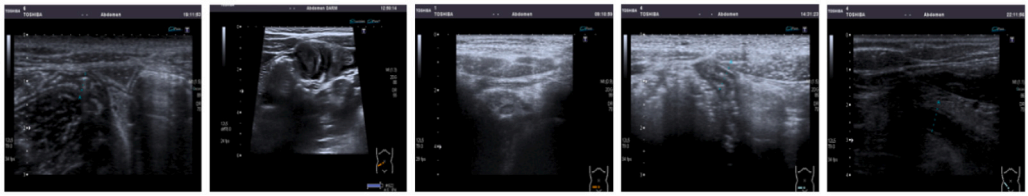
Under the corrected patient-level protocol, the framework attains a moderate discrimination (test AUC up to 0.747; cross-validated AUC ≈ 0.70), comparable to the convolutional baselines evaluated here and below the optimistic figures typically reported by image-level studies.



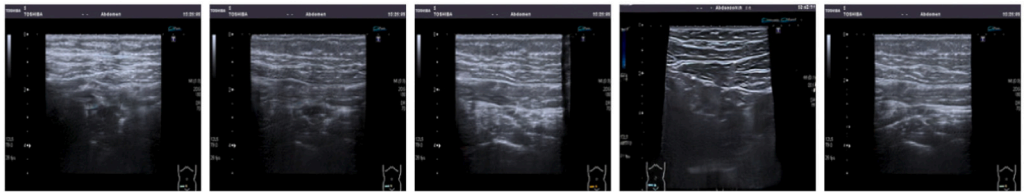
(a) Real, appendicitis.



(b) Synthetic, appendicitis.



(c) Real, non-appendicitis.



(d) Synthetic, non-appendicitis.

Fig. 5. Visual comparison of real and GAN-generated synthetic ultrasound images (representative samples; one row of five per category).

We consider this more conservative estimate a more faithful reflection of the genuine difficulty of image-only pediatric appendicitis classification. Notably, the dataset's original studies achieve stronger performance precisely by combining ultrasound with clinical and laboratory variables [8,10], which supports multimodal fusion as a promising route to higher accuracy. The present contribution is therefore methodological as much as architectural: to our knowledge this is the first study to combine class-specific GAN augmentation with a hierarchical Vision Transformer in pediatric appendicitis (Table 1), evaluated under a leakage-free, statistically grounded protocol.

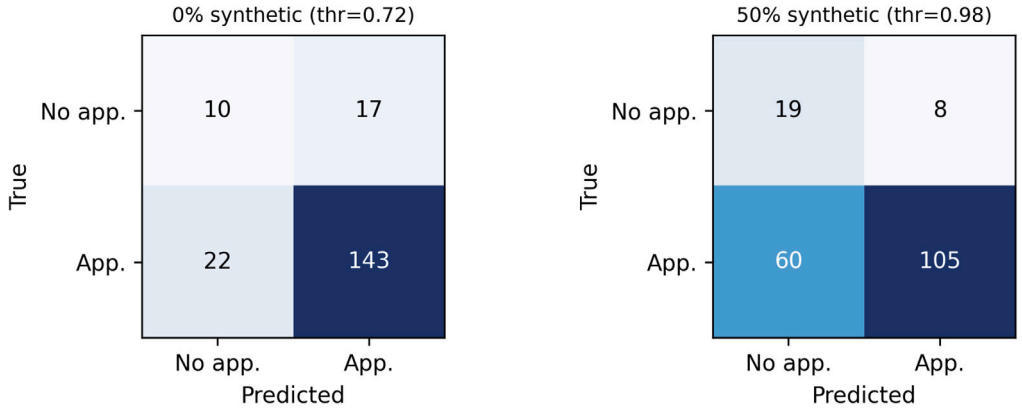
5.3. Clinical implications

From a practical standpoint, the proposed framework presents several characteristics relevant for clinical adoption. The use of synthetic data generation eliminates the need for additional real patient data, addressing privacy and consent constraints while enabling model training in institutions with limited imaging databases. The Swin Transformer architecture, in its tiny variant (`swin_tiny_patch4_window7_224`), represents a computationally feasible choice that could potentially run on standard clinical workstations equipped with a single GPU.

While these characteristics are practically attractive, the present single-center evidence – and the moderate discrimination obtained under the leakage-free protocol – do not yet support claims of clinical deployment, autonomous triage, or reductions in CT usage. We therefore frame the proposed framework as a research-stage *decision-support* concept rather than a deployable tool: its clinical utility would need to be established through external, multicenter and ideally prospective validation, and through an explicitly chosen operating point that balances sensitivity and specificity according to clinical priorities [7,49].

5.4. Limitations

Several limitations should be acknowledged. First, all experiments rely on a single publicly available, single-center dataset [12]; external,



(a) 0%: TP=143, FN=22, FP=17, TN=10.

(b) 50%: TP=105, FN=60, FP=8, TN=19.

(c) 100%: TP=110, FN=55, FP=6, TN=21.

(d) 200%: TP=111, FN=54, FP=7, TN=20.

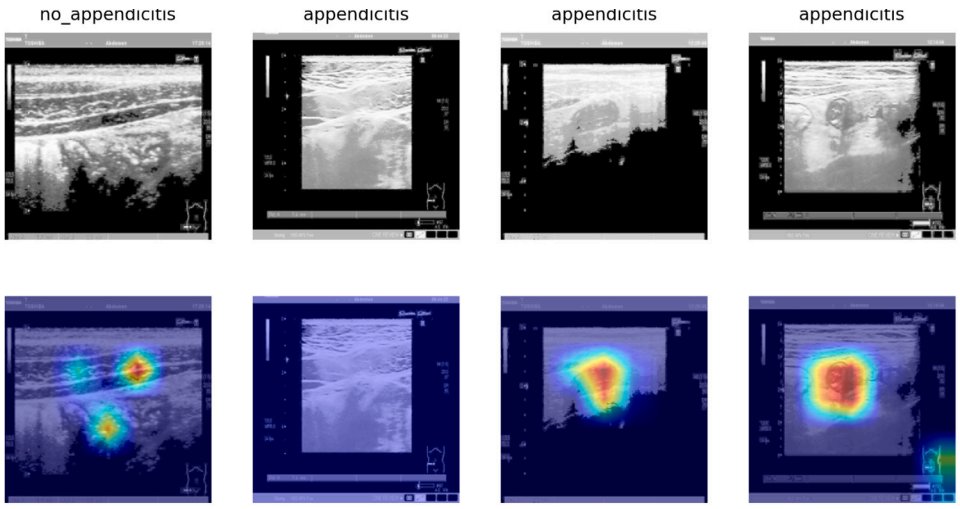
Fig. 6. Confusion matrices of the Swin Transformer on the held-out test set (165 appendicitis, 27 non-appendicitis) at the validation-selected operating point.

multicenter validation remains necessary to confirm generalizability across ultrasound devices, acquisition protocols and patient populations, with the associated domain-shift challenges discussed above. Second, the non-appendicitis class is markedly smaller (382 vs. 1339 images); although class balancing is now confined to the training set, the limited intrinsic diversity of the minority class may still constrain learning. Third, image-only classification of pediatric appendicitis is inherently difficult – as reflected by the moderate AUC obtained once data leakage is removed – and multimodal approaches that combine ultrasound with clinical and laboratory variables (as in the dataset's original studies [8,10]) constitute a promising avenue to raise performance. Fourth, downsampling to 224×224 may discard fine, millimeter-scale features; while this resolution is imposed by the ImageNet-pretrained backbones (the generator itself operates at 256×256), training on higher-resolution crops of the region of interest is left for future work. Finally, although we adopt StyleGAN2-ADA, alternative generative models such as diffusion-based approaches [19,50] were not investigated and could further improve augmentation quality.

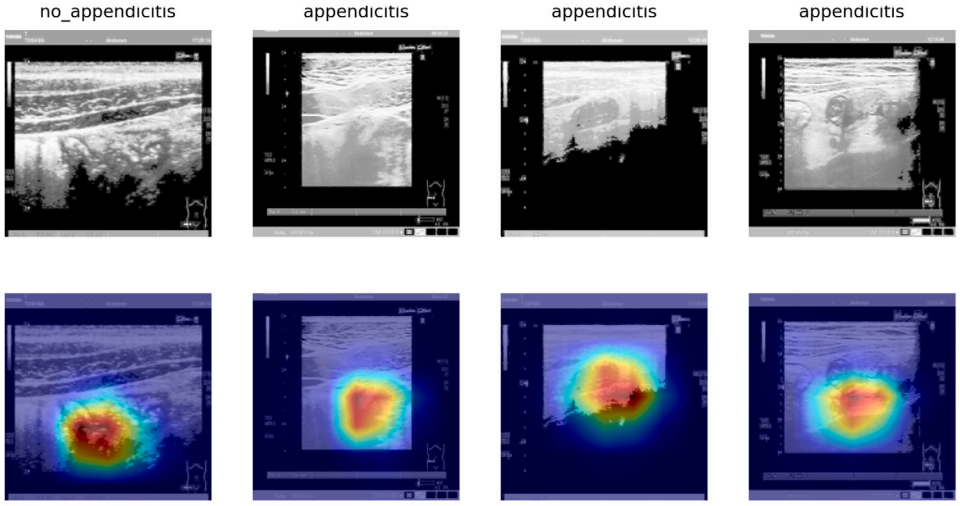
6. Conclusions

This study developed and evaluated an automated framework for pediatric appendicitis diagnosis from ultrasound images, combining GAN-based synthetic data generation with Vision Transformer-based classification under a rigorous patient-level evaluation protocol. The key findings can be summarized as follows. First, patient-level evaluation is essential in this domain: our original image-level split overestimated performance by roughly 8–12 percentage points, and once leakage is removed the task proves moderately difficult (test AUC up to 0.747; cross-validated AUC ≈ 0.70). Second, the hierarchical Vision Transformer is competitive with – but not statistically superior to – strong convolutional baselines (ResNet50, DenseNet121, EfficientNet-B0, ConvNeXt-Tiny). Third, GAN-based synthetic augmentation does not improve discrimination (AUC is essentially flat across 0%–200% synthetic data), but it yields more balanced predictions – higher specificity, balanced accuracy and MCC – with the best balance around 100% synthetic data. Fourth, the synthetic images are visually realistic and non-memorized (nearest-neighbor analysis) yet remain machine-distinguishable from real ultrasound, consistent with the absence of an AUC gain.

In summary, controlled GAN-based augmentation is best understood here as a means to *rebalance* an imbalanced classifier rather than as a source of additional discriminative power. Future work should pursue external and multicenter validation, multimodal fusion of ultrasound with clinical and laboratory data, higher-resolution region-of-interest modeling, diffusion-based generative models, and prospective clinical evaluation.



(a) Swin Transformer (attention rollout).



(b) ResNet50 (Grad-CAM).

Fig. 7. Explainability maps on held-out test images (top of each panel: input; bottom: activation overlay).

CRediT authorship contribution statement

Carlos Luis Fernández Santana: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Gerardo Fernández Álvarez:** Investigation, Formal analysis. **Ricardo S. Alonso:** Validation, Supervision, Investigation, Writing – review & editing. **Pablo Negre:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The Regensburg Pediatric Appendicitis Dataset is publicly available at the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/938/regensburg+pediatric+appendicitis) [12]. The code and processed datasets used to generate the results are available from the corresponding author upon reasonable request. No human subjects were recruited for this study, and the dataset was used in accordance with its public license terms; original data collection at the Children's Hospital St. Hedwig in Regensburg followed the institutional ethics protocols described by the dataset providers.

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